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GitHub Repo Link: <https://github.com/Traorasp/TheRootsOfWar>

The Roots of War: Predicting Organized Violence

1. Introduction & Motivation

For our final Machine Learning project, we predicted the occurrence and severity of armed conflicts using predictors based on various theories of war. Especially given the increasing violence and chaos of the international stage in the last few years, we feel it is a topical and crucial question, especially because these results can be used to try to prevent those factors from occurring, and thus reduce the frequency and severity of war. It would be naive and scientific to think that we could derive a completely reliable “social physics” that would let us predict geopolitical events, but we see great promise in finding influential factors that reduce the likelihood of catastrophic war. NGOs, responsible state actors, and intergovernmental associations like the United Nations are committed to reducing global conflict, so this kind of information could help them direct their efforts most efficiently.

We’re indebted to the war and peace studies field as a whole, which for decades has attempted to quantify the features related to conflict and document them across the world. We owe a lot specifically to the Correlates of War (CoW) project, and especially its director Stuart Bremer’s 1992 paper *Dangerous Dyads: Conditions Affecting the Likelihood of Interstate War, 1816–1965*.¹ Using CoW’s massive dataset of international conflicts, Bremer used statistical analysis to determine the most significant associated factors of war. Our mission was a similar one – we sought to use machine learning to predict the occurrence and also *severity* of wars from even more factors.

1.1 Research Questions

- **RQ1:** Can factors of wars be used to reliably predict the occurrence of conflict?
- **RQ2:** How well can these same factors predict the severity of intrastate conflict?

1.2 Expectations

We believe our theorized correlates of war will serve as reliable predictors of conflict. Although due to the complexity of geopolitical factors and the difficulty of collecting reliable data during times of conflict it will be difficult to predict the severity of intrastate conflict accurately. The severity of conflict relies on factors that are difficult to quantify or collect, so we expect only a partial explanation of the variance.

¹ Bremer, S. A. (1992). Dangerous Dyads: Conditions Affecting the Likelihood of Interstate War, 1816-1965: Conditions Affecting the Likelihood of Interstate War, 1816-1965. *Journal of Conflict Resolution*, 36(2), 309-341.

2. Data

2.1 Data Sources & Collection

We primarily used the Uppsala Conflict Data Program's excellent data to do our analysis. This organization – which began being collected in the late 1980s to help the expanding field of conflict studies – has published about casualties from all varieties of organized violence, from large-scale interstate wars to "minor" incidents and flare-ups, often geotagged to specific villages and timestamps. They collect information "in real time" by doing thorough scanning of articles from over 30,000 news organizations to identify unique events and their casualties. As we were interested in the big picture view, we chose to use their *Country-Year Dataset on Organized Violence within Country Borders*, which – as the name implies – aggregates all incidents by country and by year, which makes it easy to compare to other datasets with the same organizing principles.

To bring in the factors that we predicted would be relevant to wars, we drew from a variety of sources in line with different ideological theories of global politics: Correlates of War's *Militarized Interstate Disputes* (MID) and *National Military Capabilities* (NMC) datasets allowed us to identify the parties to conflicts and their levels of militarization respectively – crucial to Realist geopolitical understandings of conflict. The V-Dem *Democracy Index* provided scholarly democratic evaluation of all states from the present day all the way back to the 1700s, which play a large role in Liberal theories of war. The United States Department of Agriculture's *International Macroeconomic Data Set* provided historical GDP data, and the World Bank's *List of total natural resources rents (as percent of GDP)* provided economic information (both from 1970 to the present day) which is seen as key in Marxist analyses of war. With all these data points, we saw an opportunity to quantitatively analyze the ways different ideologies understand conflict.

2.2 Data Preparation & Cleaning

We did not use *all* of this data – if we did, we would have hundreds and hundreds of redundant variables. As we do not care about hyper-specific aspects of democracy and military capabilities – and we trust the previous researchers to aggregate reliable indices in a sensible way – we began by cleaning out all but the most high-level assessments from the V-Dem and NCM datasets. For the MID and UCDP datasets, we similarly removed extraneous divisions into different types of conflicts, as we felt it was most important to reduce conflict and suffering overall. Since many of the datasets used different systems of country codes, we used Vincent Arel-Bundock's *countrycode* package to translate between them to properly aggregate them into one giant, cleaned table.

We then pre-processed our data. We restricted data to only after 1989, since that is when the UCDP data source (which contains our target variables) begins. We then dropped all NA rows since numerous rows contained majority-NA values. We eventually narrowed our scope to a total of 31 predictor variables, displayed below.

2.3 Predictor Variables

Variable name	Type	Description
ccode	integer	Correlates of War country code.
year	integer	Year of observation.
cinc	float	Composite Index of National Capability (CINC) score.
region_cy	string	Region where the country is located.
sb_exist_cy	bool	The existence of state-based violence within the borders of a country in a given year. It is 1 if state-based violence occurs.
sb_dyad_count_cy	integer	The number of state-based dyads engaging in organized violence within the borders of a country in a given year.
sb_dyad_ids_cy	list of integers	The names of state-based dyads engaging in organized violence within the borders of a country in a given year. NO_DYAD is coded when there was no relevant record in UCDP GED.
sb_intrastate_exist_cy	bool	The existence of intrastate state-based violence within the borders of a country in a given year. It is 1 if intrastate state-based violence occurs.
sb_intrastate_dyad_count_cy	integer	The number of intrastate state-based dyads engaging in organized violence within the borders of a country in a given year.
sb_intrastate_dyad_ids_cy	list of integers	The IDs of intrastate state-based dyads engaging in organized violence within the borders of a country in a given year. NO_DYAD is coded when there was no relevant record in UCDP GED.
sb_intrastate_main_govt_inv_inc_omp_cy	bool	The involvement of the government of the country in forming the incompatibility of at least one intrastate state-based violence within the borders of a country in a given year. It is 1 if the government of the country is among the parties forming the incompatibility of at least one intrastate state-based violence.
sb_interstate_exist_cy	bool	The existence of interstate state-based violence within the borders of a country in a given year. It is 1 if interstate state-based violence occurs.

sb_interstate_dyad_count_cy	integer	The number of interstate state-based dyads engaging in organized violence within the borders of a country in a given year.
sb_interstate_dyad_ids_cy	list of integers	The IDs of interstate state-based dyads engaging in organized violence within the borders of a country in a given year. NO_DYAD is coded when there was no relevant record in UCDP GED.
sb_interstate_main_govt_inv_inc_omp_cy	bool	The involvement of the government of the country in forming the incompatibility of at least one interstate state-based violence within the borders of a country in a given year. It is 1 if the government of the country is among the parties forming the incompatibility of at least one interstate state-based violence.
ns_exist_cy	bool	The existence of non-state violence within the borders of a country in a given year. It is 1 if non-state violence occurs.
ns_dyad_count_cy	integer	The number of non-state dyads engaging in organized violence within the borders of a country in a given year.
ns_dyad_ids_cy	list of integers	The IDs of non-state dyads engaging in organized violence within the borders of a country in a given year. NO_DYAD is coded when there was no relevant record in UCDP GED.
os_exist_cy	bool	The existence of one-sided violence within the borders of a country in a given year. It is 1 if one-sided violence occurs.
os_dyad_count_cy	integer	The number actors carrying out one-sided violence within the borders of a country in a given year.
os_dyad_ids_cy	list of integers	The IDs of one-sided incidents (dyad IDs) within the borders of a country in a given year. They are the dyad IDs stated on the UCDP Georeferenced Events Dataset. NO_DYAD is coded when there was no relevant record in UCDP GED.
os_main_govt_inv_cy	bool	The involvement of the government of the country in one-sided violence within the borders of a country in a given year. It is 1 if the government of the country is involved in one-sided violence.
os_nsgroup_inv_cy	bool	The involvement of any non-state group in one-sided violence within the borders of a country in a given year. It is 1 if a non-state group is involved in one-sided violence.

v2x_polyarchy	float	Electoral democracy index
v2x_libdem	float	Liberal democracy index
v2x_partipdem	float	Participatory democracy index
v2x_delibdem	float	Deliberative democracy index
v2x_egaldem	float	Egalitarian democracy index
gdp	float	Gross Domestic Product (adjusted for inflation)
gdp_percent_change	float	% change of Gross Domestic Product (adjusted for inflation)
nat_res_rent	float	Natural resource rent as a proportion of GDP

2.4 Target Variables

We had 4 target variables, each representing types of casualties:

Variable name	Type	Description
ctd_parties	integer	The best estimate for cumulative fatalities of <u>the parties</u> in organized violence within the borders of a country in a given year.
ctd_civ	integer	The best estimate for cumulative <u>civilian</u> fatalities in organized violence within the borders of a country in a given year.
ctd_unk	integer	The best estimate for cumulative <u>unknown</u> fatalities in organized violence within the borders of a country in a given year.
ctd_best	integer	The best estimate for <u>total</u> fatalities in organized violence within the borders of a country in a given year.

From this collection, we split our dataset into training and testing sets with a test set proportion of 0.2. We then standard-scaled our numeric variables for regularization since models like Ridge, KNN and the Neural Network rely on these ranges. We one-hot-encoded the categorical variables ccode and region_cy to properly process them. Lastly, we had a few columns that were made up of varying sized arrays of ids. In order to preserve their meaning and use them as presence or absence indicators, we multi-label-encoded them.

3. Methods

3.1 Ridge Regression

For all the models we used the same preprocessed training and test data. For each model type and target variable we used grid search and a cross validation with 5 folds to fine-tune the hyperparameters and select the best model according to their `neg_root_mean_squared_error` score. We started with a simple linear regression model whose results were very inaccurate and inconsistent. So we opted to instead use a Ridge regression model to deal with the multicollinearity that we assumed existed because of how often geopolitical variables such as democratic index, `cinc` and `gdp` are often highly correlated. The Ridge regression model was trained with varying alpha values ranging from 0.001 to 1000. We received these results:

Score	<code>ctd_parties</code>	<code>ctd_civ</code>	<code>ctd_unk</code>	<code>ctd_best</code>
RMSE	1716.637	1758.599	415.564	3393.329
R ²	0.602	-1.754	-0.590	0.219

While the Ridge model performed quite well on `ctd_parties`, alright on `ctd_best`, it performed extremely poorly on `ctd_civ` and `ctd_unk`. This shows that for `ctd_parties` and `ctd_best` variance was meaningfully predictable by the dependent variables. Meanwhile `ctd_civ` and `ctd_unk` variance were not predictable through the dependent variables. Meaning they were very likely nonlinear in nature. It was also interesting to note that the best Ridge models tended to have a high alpha value implying that a large amount of regularization was necessary. So there might be a multitude of variables that are redundant and contribute little to the models. It would be interesting to investigate the individual weights of the predictor variables on the target variables.

3.2 K-Nearest Neighbors

The next model we tried after the Ridge model was KNN. Since `ctd_civ` and `ctd_unk` were most likely nonlinear we opted to use KNN, a non-parametric model, to see if they had local relationships. For the KNN we trained models with varying k neighbors from 3 through 12, KNN weights of uniform or distance, and lastly through a feature selection of 2 to 31 variables. The reason for the feature selection was to deal with KNN suffering from high dimensionality, so we trained it with varying numbers of variables. Our results were:

Score	<code>ctd_parties</code>	<code>ctd_civ</code>	<code>ctd_unk</code>	<code>ctd_best</code>
RMSE	1844.464	503.065	272.071	3874.090
R ²	0.541	0.774	0.318	-0.0173

These results were very promising and demonstrated that `ctd_civ` and `ctd_unk` were non linear and instead more locally related. Specifically for `ctd_civ` and `ctd_unk` which KNN was the best out of any model. The best model all had 3 neighbors and used all the features, except for `ctd_civ` which used 25

features. This meant that our worries of high dimensionality causing a problem were not particularly necessary. Most data except for ctd_best had a strong local relationship to each other which, knowing this led to our next models. For further studies it would be interesting to test out even more combinations of hyperparameters. In addition, to using Principle Component Analysis (PCA) on the variables to reduce the dimensionality of the input variables.

3.3 Random Forest

In order to deal with the nonlinear data, we used a Random Forest model in the hope that it would better be able to handle complex features of the data than the Ridge model was. We trained trees with the following hyperparameters: tree depth from 3 to 12, minimum leaf sizes from 1 to 2, minimum split sizes of 2 to 5, features per split of 5 to 31, and number of trees from 75 to 175.

Score	ctd_parties	ctd_civ	ctd_unk	ctd_best
RMSE	1982.750	1018.843	327.418	3242.082
R ²	0.469	0.0754	0.0127	0.287

The results of the Random Forest model were overall satisfactory. There were no negative r^2 scores, meaning that most of the data indeed had some (nonlinear) relationship. More importantly, it had the best score of the previous models in measuring ctd_best. The tree model is decently fit to measure the threshold effect and nonlinearity of the data.

3.4 Gradient Boosting

Seeing the success of the Random Forest model, we then chose to use a Gradient Boosting model in order to try to better predict the data. Gradient Boosting models usually outperform Random Forests and are better at capturing non linear patterns due to their corrective nature. So we trained over the following hyperparameters: learning rate of 0.01 to 1, minimum leaf sizes of 1 to 3, minimum split sizes of 2 to 4, features per split of 5 to 31, number of trees from 75 to 175. Our results were:

Score	ctd_parties	ctd_civ	ctd_unk	ctd_best
RMSE	1895.502	2658.791	485.432	5727.098
R ²	0.515	-5.296	-1.170	-1.223

The results of the Gradient Boosting model were extremely inaccurate and aside from ctd_parties all the other target variables showed little relation. This was mostly likely due to some form of overfitting, not enough regularization, or the model being too sensitive to the noise. It serves as a good example that a more complex model isn't necessarily better. In further study it would be interesting to see what a Gradient Boosting model would look like with more regularization and a different set of hyper parameter tunings.

3.4 Neural Network

Lastly we decided to see how a Neural Network would be able to deal with the data. For the Neural Network we turned all the data into tensors and split the training data into a training and evaluation data set. Then using a parameter grid we trained different models with different parameters. The hyper parameters were 1st hidden layer neurons 32 to 256, 2nd hidden layer neurons 32 to 256, dropout rates of 0 to 0.6, learning rate 0.0005 and 0.001, and batch_size of 32 or 64. Our model had 31 inputs for each predictor variable and 4 outputs, one for each target variable. This allows the model to learn relations between the 4 target variables which we know are closely related. We used a simple Relu activation function since we know output should never be negative. We also used batch normalization and dropout for regularization, and optimizing learning rate. For the optimizer we used Adam as a safe and stable choice for the regression task. Lastly instead of the MSE loss function we used the SmoothL1Loss function to stabilize the model to deal with outliers and exploding gradients. It is also very similar to the MSE function for small errors. Our results were:

Score	ctd_parties	ctd_civ	ctd_unk	ctd_best
RMSE	2371.691	922.627	303.439	3137.533
R ²	0.241	0.241	0.152	0.332

For ctd_best, the Neural Network model performed the best compared to any other model, showing that it had a complex nonlinear relationship. This might also be due to how ctd_best is calculated, it represents the best estimate of casualties so it would make sense that it is correlated to the other three target variables. With the Neural Network model being the only model to learn patterns between the 4 target variables. This might explain why it performed the best, although it would require further testing. Researchers could train more models taking this into account and or find the correlation between the 4 target variables. As for the other target variables it performed decently well on all of them. This might imply that they are a lot more linear and or locally related than expected.

4. Results

4.1 Summary

We trained multiple types of models with grid search cross-validation, fine-tuning the hyperparameters to select the best model. For each of our target variables we trained a model using our 31 predictor variables. The best results of our analysis for each of our target variables were:

Target	Best Model	RMSE	R ²
ctd_parties	Ridge	1716.637	0.602
ctd_civ	KNN	503.065	0.774
ctd_unk	KNN	272.071	0.318
ctd_best	Neural Network	3137.533	0.332

4.2 Explanation

The ctd_parties was very well estimated by a Ridge model meaning that there was a moderately strong linear correlation between the estimators and the target variable. For ctd_civ was very well predicted by the KNN model, implying that the target variable is likely non linear and instead has a very local relationship. For ctd_unk the KNN model was able to moderately predict it meaning the target variable has some local relationship, but is most likely some other form of non linear relation. It is not surprising that the unknown casualties have a very non linear and complex relation. Lastly ctd_best was decently predicted by the Neural Network model showing that it has some form of non linear relationship and correlation to the other target variables. Since the Neural Network model was the best able to predict it and it was the only one taking into account relations between the 4 target variables.

5. Conclusion

5.1 Positive Results

Our models support our research hypothesis that it is possible to predict the military casualties of intrastate conflict, with relatively strong accuracy. Specially for the most important target variables of civilian and involved parties casualties. Our Ridge model using the predictor variables was able to predict 60% of the variance in ctd_parties, which considering the many complex factors of geopolitical conflicts is very good for a linear model. This helps us understand that ctd_parties likely has a strong linear correlation and can be predicted using such models. In the future different types of linear models can be trained to more accurately predict involved parties' casualties for intra state conflict. For our KNN model we were able to do even better and predict 77% of the variance in ctd_civ using the predictor variables. This informs us of a strong local relationship between civilian casualties. Taking this into account we can use conflicts with similar conditions to more accurately predict the civilian casualties for intrastate conflict. In other words, we have found factors that can presumably be successfully deployed to partially predict the occurrence severity of conflicts, which was our goal.

5.2 Non-Results & Limitations

It makes sense that we could not explain *all* of the variance – geopolitics and war are extremely complex, contingent social systems that can never be fully predicted. Every conflict and historical event is unique and irreducible to quantitative variables. In fact, this leads us to be tentative and skeptical of even our apparently promising results. Given that we do not have *that* many data points, there is a danger of overfitting or "cheating" the data. GDP being an element in our predictors may be fraught – while it's likely true that poorer countries are more likely to have armed conflict happen on their territory, it is definitely true that war on the territory of a country *causes* that country to become poorer! We may be gaining useless information by reversing causation.

Additionally, our analysis may suffer from our failure to account for attributes of other involved countries in a conflict besides the one whose territory it is on. To compare a concrete example, one can't explain the Russo-Ukrainian War through only looking at the Ukrainian state; it's obviously more important to look at the expansionist regime of Vladimir Putin in Russia. The way our algorithm currently works has an almost "victim-blaming" bias to it. We must also recognize that the period from 1989 to the present is quite a short and historically odd time in history. It may have been a better idea to choose a dataset like MID as our base that went back centuries, even given the loss of precision in severity of any particular conflict. This would have given us a greater range of kinds of states and allowed for a bigger-picture philosophical look at why wars happen.

The post-1989 timespan leads to disproportionalities in the data – for example, heavily protracted civil wars in Syria and Iraq sway the data toward assuming natural resource-rich Middle Eastern countries are most prone to war, when this may not be totally true in the long-term. Despite these many concerns, the literature suggests that deep learning may enable researchers to predict what were previously seen as "random" outbreaks.² Determining what is useful and what is just overfitting will require a fine-tuned, skeptical approach.

5.2 Ethical Issues & Final Remarks

It is also worth examining the ethical issues of this kind of research. Predictions of war do not exist in a vacuum. They may inform decisions of organizations to *disinvest* from states that are vulnerable to conflict, and thus play into a vicious cycle where victims of war are deprioritized more than they already are. It is also worth seriously considering the implications this has on power relations themselves.³ The last thing we'd want to do with research is to inspire fatalism – the point of this research would be to enable preventative action to spread peace, but we would need to be careful to ensure this. A noticeable aspect of conflict that is missing from our research is just that – the complex, dutiful work of peace negotiators that brings an end to war and stops it from starting as much as possible. To publish a study like this, we would need to do more self-analysis to ensure we truly are creating reliable data that can be employed in such a way: to prevent wars when possible, and mitigate the harm and save lives if not.

² Murphy, Max, Ezra Sharpe, and Kayla Huang. "The Promise of Machine Learning in Violent Conflict Forecasting." *Data & Policy* 6 (2024): e35. <https://doi.org/10.1017/dap.2024.27>.

³ For more details, see Tshilidzi Marwala. "Can Deep Learning Predict War, and Should It?," United Nations University, UNU Centre, 2026-04-20, <https://unu.edu/article/can-deep-learning-predict-war-and-should-it>.

5.3 Afterthoughts

If we were to redo this project from the beginning, we would start by selecting more appropriate data and spend less time trying to struggle with the MID dataset, which turned out to be unusable for our purposes. If possible, we would collect a wider set of predictor variables to increase the models' capabilities and explore more factors. We would also make more data visualizations to get a better understanding of the data and make it more comprehensible to readers.. For the models we would focus on polishing the Ridge, Random Forest and KNN models rather than exploring other methods that ended up as dead-ends. We would try a wider set of hyper parameters and metrics to infer more from our models.

Generally, we feel these approaches could have yielded more productive results. Still, we believe that our current data contributes to the greater cause for peace.